# Brian Chu

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## WORK EXPERIENCE

**Scale AI** | San Francisco, CA **April 2026 - Present**

*Forward Deployed AI Engineer*

- Shipped 20+ structured-output features to enterprise health clinic and engineering firm clients by building internal MCP tooling with a deterministic architecture that plugs into AI agents and evals
- Recovered $10M in credits across 15K+ documents by deploying an AI extraction pipeline orchestrated in Temporal, routing low-confidence extractions to human-in-the-loop (HITL) reviewers
- Informed a $2M budget reallocation by building executive metrics dashboards on 1M+ weekly logged events, replacing manual reports with real-time visibility into credit recovery

*Software Engineer*

- Cut new feature build time from 2 weeks to 3 days by reusing shared, deterministic MCP tooling with Redis caching across client projects instead of building each integration from scratch
- Caught agent regressions before client releases by scheduling 1,000+ daily eval cases against AI agent outputs with GCP Cloud Run jobs and Cloud Scheduler for enterprise deployments
- Automated trade research with a multi-agent portfolio trading system of tool-calling research and strategy agents, gated by deterministic risk checks, AWS IAM-scoped access, and HITL approval

**Stealth Startup** | Remote **May 2022 - April 2026**

*AI Engineer*

- Replaced manual machinist quoting with autonomous pricing across 7 cost categories by deploying tool-calling agent loops over 3 LLM providers with Pydantic output via FastAPI microservices
- Achieved 40% higher engineering drawing analysis accuracy by developing a VLM-based preprocessing pipeline using Gemini Vision for classification, assembly trees, and BOM extraction
- Cut pricing lookup latency 70% by consolidating three quote workflows into unified React, FastAPI, and PostgreSQL microservices with recursive BOM quoting and cached OAuth vendor pricing APIs

**Jacobs School of Engineering, UCSD** | La Jolla, CA **January 2023 - June 2023**

*Thermal Process Engineer*

- Achieved 30% higher energy efficiency and 95% methanol conversion by designing an industrial-scale dimethylamine reactor, heat exchanger system, and P&ID in Aspen Plus & COMSOL

## PROJECTS

**Speed Demon - F1 AI Racing Technology Automation**

- Engineered real-time telemetry system using OpenTelemetry, with a time-series database on AWS EC2, integrating LangChain agents utilizing skills and MCP with GraphRAG to predict driver style
- Orchestrated a distributed data processing backend by implementing batch and stream processing using PySpark on AWS EC2 cloud infrastructure with centralized monitoring

**NLP Drug-Drug Interaction Prediction System**

- Predicted combined drug effects with 82% accuracy by developing a Knowledge Graph Neural Network for 10K+ drug-drug interaction descriptions with TensorFlow and Neo4j
- Processed pharmacological text data by tokenizing with spaCy and building feature pipelines with Hugging Face Transformers, improving model input quality for the Knowledge Graph Neural Network

## EDUCATION

**University of California, San Diego** | San Diego, CA

*Master of Science in Chemical Engineering, FE Chemical*

*Bachelor of Science in Computer Science and Chemical Engineering*

## SKILLS

**Languages | Frameworks:** Python | TypeScript | C++ | SQL | LangChain | TensorFlow | FastAPI | Pydantic

**Tools | Infrastructure:** AWS | GCP | Kubernetes | Docker | Temporal | PostgreSQL | Redis | CI/CD